

THE PRODUCT RULE

If $p(x) = f(x)g(x)$, then $p'(x) = f'(x)g(x) + f(x)g'(x)$.

Note: Leibniz Notation: If u and v are functions of x , $\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}$

Proof:

Let $p(x) = f(x)g(x)$, then

$$p'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{(x+h) - x}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) + 0 - f(x)g(x)}{h}$$

Just like multiplying by 1 is a powerful tool in math, so is adding 0
In this case, $0 = -f(x)g(x+h) + f(x)g(x+h)$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}$$

Factor out $g(x+h)$ from the first two terms of the numerator, and
we'll factor out $f(x)$ from the last two terms of the numerator

$$= \lim_{h \rightarrow 0} \left\{ \left[\frac{f(x+h) - f(x)}{h} \right] g(x+h) + f(x) \left[\frac{g(x+h) - g(x)}{h} \right] \right\}$$

$$= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] \lim_{h \rightarrow 0} g(x+h) + \lim_{h \rightarrow 0} f(x) \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right]$$

$$= f'(x)g(x) + f(x)g'(x)$$

Examples

1. Differentiate $h(x) = (x^3 - 2x)(3x^4 + 2x + 8)$ using the product rule.

Degree = 7

$$\begin{aligned}
 h'(x) &= \underbrace{(x^3 - 2x)}_{\text{Degree} = 6} \cdot (3x^4 + 2x + 8) + (x^3 - 2x) \cdot (3x^4 + 2x + 8)' \\
 &= (3x^2 - 2)(3x^4 + 2x + 8) + (x^3 - 2x)(12x^3 + 2) \\
 &= 9x^6 + 6x^3 + 24x^2 - 6x^4 - 4x - 16 + 12x^6 + 2x^3 - 24x^4 - 4x \\
 &= 21x^6 - 30x^4 + 8x^3 + 24x^2 - 8x - 16
 \end{aligned}$$

2. Determine the value of $f'(-1)$ for the function $f(x) = (3x^4 - 12x^2 + 4x - 9)(6x^7 - 4x^4 + 18)$.

$$f'(x) = (12x^3 - 24x + 4)(6x^7 - 4x^4 + 18) + (3x^4 - 12x^2 + 4x - 9)(42x^6 - 16x^3)$$

$$f'(-1) = (-12 + 24 + 4)(-6 - 4 + 18) + (3 - 12 - 4 - 9)(42 + 16)$$

$$f'(-1) = (16)(8) + (-22)(58)$$

$$f'(-1) = 128 - 1276$$

$$f'(-1) = -1148$$

Determine an expression for $p'(x)$ if $p(x) = f(x)g(x)h(x)$.

$$p'(x) = \underbrace{f'(x)}_{(2x^2+5x)} \underbrace{g(x)}_{(x-1)} \underbrace{h(x)}_{(2x^2+5x)} + \underbrace{f(x)}_{(2x^2+5x)} \underbrace{g'(x)}_{(1)} \underbrace{h(x)}_{(2x^2+5x)} + \underbrace{f(x)}_{(2x^2+5x)} \underbrace{g(x)}_{(x-1)} \underbrace{h'(x)}_{(4x+5)}$$

This is called the **extended product rule** for three functions.

3. Differentiate the cubic function $f(x) = x(2x+5)(x-1)$ by using the extended product rule.

$f(x) = 2x^3 + 5x^2 - 5x$
 $f'(x) = 6x^2 + 10x - 5$

$$f'(x) = (1)(2x+5)(x-1) + x(2)(x-1) + x(2x+5)(1)$$

$$f'(x) = (2x+5)(x-1) + 2x(x-1) + x(2x+5)$$

$$= 2x^2 + 3x - 5 + 2x^2 - 2x + 2x^2 + 5x$$

$$= 6x^2 + 6x - 5$$

$$h(x)$$

4. If $g(x) = x^2 f(x)$, $f(2) = -2$ and $g'(2) = 8$, then determine $f'(2)$.

$$g'(x) = (x^2)' f(x) + x^2 f'(x)$$

$$g'(x) = 2x f(x) + x^2 f'(x)$$

$$g'(2) = 2(2) f(2) + (2)^2 f'(2)$$

$$8 = 4(-2) + 4 f'(2)$$

$$16 = 4 f'(2)$$

$$\therefore f'(2) = 4$$

Product rule

THE QUOTIENT RULE

If $h(x) = \frac{f(x)}{g(x)}$, then $h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$, $g(x) \neq 0$.

Note: Leibniz Notation:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{\frac{du}{dx} v - u \frac{dv}{dx}}{v^2}$$

Proof:

Let $h(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$, therefore

$$h(x)g(x) = f(x)$$

$$h'(x)g(x) + h(x)g'(x) = f'(x)$$

product rule

$$h'(x)g(x) = f'(x) - h(x)g'(x)$$

$$h'(x) = \frac{f'(x) - h(x)g'(x)}{g(x)}$$

$$= \frac{f'(x) - \frac{f(x)}{g(x)}g'(x)}{g(x)}$$

$$= \frac{f'(x) - \frac{f(x)}{\cancel{g(x)}}g'(x)}{g(x)} \times \frac{\cancel{g(x)}}{g(x)}$$

$$= \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Examples

→ $(3x-4)(x^2+5)^{-1}$ product rule

1. Determine the derivative of $h(x) = \frac{3x-4}{x^2+5}$.

$$h'(x) = \frac{(3x-4)'(x^2+5) - (3x-4)(x^2+5)'}{(x^2+5)^2}$$

$$h'(x) = \frac{3(x^2+5) - (3x-4)(2x)}{(x^2+5)^2}$$

$$h'(x) = \frac{3x^2 + 15 - (6x^2 - 8x)}{(x^2+5)^2}$$

$$h'(x) = \frac{-3x^2 + 8x + 15}{(x^2+5)^2}$$

wolfram alpha.com

2. Determine the coordinates of each point on the graph of $f(x) = \frac{2x+8}{\sqrt{x}}$ where the tangent is

horizontal. horizontal tangent $\Rightarrow m_t = 0$

$$f'(x) = \frac{(2\sqrt{x})' - (2x+8)(\sqrt{x})'}{(\sqrt{x})^2}$$

$$f'(x) = \frac{4x - 2x - 8}{2\sqrt{x^3}}$$

$$f'(x) = \frac{2x-8}{2\sqrt{x^3}} = \frac{x-4}{\sqrt{x^3}}$$

Set $f'(x) = 0$

$$\frac{x-4}{\sqrt{x^3}} = 0 \Rightarrow x-4=0$$

$$x-4=0$$

$$x=4 \Rightarrow y=8$$

$$\therefore \text{pt @ } (4, 8)$$

$$f(4) = 8$$

$$f(x) = \frac{2x}{\sqrt{x}} + \frac{8}{\sqrt{x}} = 2\sqrt{x} + 8x^{-\frac{1}{2}}$$

$$\begin{aligned} \frac{2x}{\sqrt{x}} &\rightarrow \frac{2\sqrt{x} \cdot \sqrt{x}}{\sqrt{x}} \rightarrow 2\sqrt{x} \cdot 1 = 2\sqrt{x} \\ \frac{8}{\sqrt{x}} &\rightarrow \frac{8x^{\frac{1}{2}}}{x^{\frac{1}{2}}} \rightarrow 8x^{-\frac{1}{2}} = 8x^{-\frac{1}{2}} \end{aligned}$$

Additional Questions

1. Determine the equation of the tangent line to $g(x) = \left(\frac{1}{x^3} + 1\right)(x-1)$ at $x = -1$.

$$g'(x) = \left(-\frac{3}{x^4}\right)(x-1) + \left(\frac{1}{x^3} + 1\right)$$

$$g'(-1) = (-3)(-2) + 0$$

$$g'(-1) = 6$$

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 6(x + 1)$$

$$y = 6(x + 1)$$

2. If $g(x) = \frac{f(x)}{\sqrt{x-1}}$, where $f(5) = 8$, and $f'(5) = -5$, find $g'(5)$.

$$g'(x) = \frac{f'(x)(\sqrt{x-1}) - f(x) \cdot \frac{1}{2\sqrt{x-1}}}{x-1}$$

$$g'(5) = \frac{f'(5)\sqrt{4} - f(5)}{2\sqrt{4}}$$

$$g'(5) = \frac{(-5)(2) - 8}{4}$$

$$g'(5) = -3$$

3. Determine the points on the curve $y = \frac{x}{x+1}$ where the **normal** line is parallel to $x + y = 2$.

$$x + y = 2$$

$$y = -x + 2 \Rightarrow \text{normal is parallel} \therefore m_t = 1$$

$$\frac{dy}{dx} = \frac{(1)(x+1) - (x)(1)}{(x+1)^2}$$

$$\frac{dy}{dx} = \frac{1}{(x+1)^2}$$

$$\frac{1}{(x+1)^2} = 1$$

$$(x+1)^2 = 1$$

$$x+1 = \pm 1$$

$$x = 0, -2$$

$\therefore$ points are

$$(0, 0)$$

&

$$(-2, 2)$$